

BUKOKA, Tanzania, United Republic of

WMO: 637290

Lat: 1.3328S

Lon: 31.8214E

Elev: 1137

StdP: 88.39

Time Zone: 3.00 (E03)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	16.8	17.2	12.8	10.6	20.1	13.8	11.3	20.0	7.4	24.2	5.4	23.3	0.5	240	0.209

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.1	28.2	21.8	27.9	21.7	27.2	21.5	23.0	27.1	22.6	26.7	22.2	26.3	2.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	18.8	25.9	21.2	18.2	25.4	20.9	17.9	25.2	74.5	27.2	72.9	26.8	71.4	26.3	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 6.8	5.4	4.8	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	13.1	25.2	1.0	0.9	12.4	25.8	11.8	26.3	11.3	26.8	10.5	27.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.2	22.5	22.9	22.6	22.3	22.1	22.1	21.8	21.9	22.0	22.3	22.2	22.1		
	DBStd	0.94	1.02	1.09	1.12	0.87	0.81	0.73	0.72	0.73	0.83	0.83	0.92	0.97		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	4462	387	361	389	369	376	364	364	368	360	382	366	375		
	CDD18.3	1421	128	128	131	119	118	114	106	110	110	123	116	117		
	CDH23.3	5639	599	688	606	419	394	463	421	412	409	447	387	393		
	CDH26.7	305	57	89	65	14	4	6	6	6	13	17	12	18		
(14) Wind	WSAvg	1.2	0.8	1.1	1.0	1.0	1.3	1.6	1.6	1.5	1.2	1.1	0.9	0.8		
(15) Precipitation	PrecAvg	2141	156	167	253	375	320	81	52	70	100	168	205	196		
	PrecMax	2751	286	315	408	630	523	257	182	141	222	372	478	362		
	PrecMin	1662	48	48	86	117	141	4	0	8	37	80	103	48		
	PrecStd	271	60	78	59	107	100	48	51	35	49	79	84	83		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.9	29.1	29.1	27.9	27.1	27.2	27.2	27.2	27.8	28.0	27.8	28.1		
		MCWB	22.0	21.9	22.4	22.1	21.5	20.7	20.5	21.2	21.3	21.2	21.3	20.8		
	2%	DB	28.0	28.2	28.1	27.0	26.4	26.7	26.6	26.7	27.0	27.1	27.0	27.1		
		MCWB	21.8	21.9	22.1	21.9	21.4	20.8	20.2	21.0	21.3	21.4	21.4	21.2		
	5%	DB	27.2	27.9	27.4	26.2	26.0	26.1	26.0	26.1	26.2	26.4	26.2	26.4		
		MCWB	21.5	21.9	21.8	21.6	21.2	20.5	20.0	20.6	20.9	21.2	21.2	21.2		
	10%	DB	26.6	27.1	26.8	25.8	25.4	25.8	25.4	25.4	25.8	25.9	25.7	25.8		
		MCWB	21.3	21.7	21.6	21.4	21.0	20.4	19.7	20.2	20.8	20.9	21.1	21.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.4	23.4	23.4	22.9	22.7	22.2	22.1	22.7	22.7	23.4	22.7	22.6		
		MCDB	27.5	27.8	27.8	26.7	26.0	26.0	25.9	26.1	26.6	26.4	26.5	26.3		
	2%	WB	22.5	22.7	22.7	22.4	22.1	21.6	21.3	21.7	22.0	22.2	22.1	22.0		
		MCDB	26.8	27.2	27.2	26.2	25.6	25.3	25.7	26.1	26.0	26.0	25.9	26.0		
	5%	WB	22.0	22.2	22.2	22.0	21.7	21.1	20.8	21.1	21.5	21.6	21.6	21.5		
		MCDB	26.3	26.8	26.7	25.7	25.2	25.3	24.9	25.2	25.5	25.5	25.4	25.4		
	10%	WB	21.6	21.8	21.7	21.6	21.3	20.6	20.4	20.6	21.0	21.1	21.2	21.1		
		MCDB	25.7	26.3	26.1	25.1	24.7	24.8	24.5	24.6	25.0	25.0	25.0	24.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.1	7.7	6.6	6.5	7.1	7.8	7.4	7.5	7.2	7.1	7.1		
		MCDBR	9.2	9.3	9.1	7.6	7.4	8.0	8.7	8.4	8.6	8.2	8.0	8.2		
	5% WB	MCWBR	5.3	5.7	5.5	4.2	4.1	4.3	4.7	4.8	5.1	4.5	4.3	4.5		
		MCWBR	8.6	8.7	8.5	7.1	7.0	7.5	7.8	7.8	8.2	7.6	7.6	7.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.436	0.437	0.411	0.362	0.360	0.396	0.427	0.432	0.395	0.384	0.373	0.408		
		taud	2.306	2.312	2.430	2.597	2.591	2.436	2.304	2.297	2.416	2.453	2.517	2.404		
	Ebn at Noon	Ebn at Noon	896	898	914	937	916	868	845	861	914	937	951	919		
		Edh at Noon	137	137	121	100	96	110	127	133	121	118	110	123		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.87	5.36	5.09	4.89	4.83	5.06	5.21	5.02	4.89	4.83	4.50	4.49		
	RadStd	RadStd	0.51	0.33	0.38	0.43	0.29	0.45	0.36	0.40	0.35	0.35	0.28	0.39		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page